

ANALYYSI I: LIMITS

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ABSTRACT. Notes for the University of Helsinki course MAT11003, taught in Finnish from 01.08.2026–17.10.2026.

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1. COURSE INFORMATION

The official course information is available on the University of Helsinki course page.

2. LECTURE NOTES

2.1. 01.09.2026.

2.1.1. *The absolute value function.*

Definition 2.1. For any real number $x \in \mathbb{R}$, the *absolute value* (or *modulus*) of x , denoted by $|x|$, is defined by

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0. \end{cases}$$

Geometrically, $|x - y|$ is the distance between the points x and y on the real line.

Example 2.2. Fix $y \in \mathbb{R}$ and let $h > 0$. The inequality $|x - y| < h$ can be written in several equivalent ways:

$$\begin{aligned} |x - y| < h &\iff -h < x - y < h \\ &\iff y - h < x < y + h \\ &\iff x \in (y - h, y + h). \end{aligned}$$

Thus, $|x - y| < h$ means that x lies in the open interval of radius h centred at y .

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Example 2.3. As a concrete instance of the preceding equivalences,

$$\begin{aligned} |2x - 1| < 3 &\iff -3 < 2x - 1 < 3 \\ &\iff -2 < 2x < 4 \\ &\iff -1 < x < 2 \\ &\iff x \in (-1, 2). \end{aligned}$$

Theorem 2.4 (Triangle and reverse triangle inequalities). *For all $x, y \in \mathbb{R}$,*

$$\begin{aligned} |x + y| &\leq |x| + |y|, \\ ||x| - |y|| &\leq |x - y|. \end{aligned}$$

Proof. By definition 2.1,

$$\begin{aligned} -|x| &\leq x \leq |x|, \\ -|y| &\leq y \leq |y|. \end{aligned}$$

Adding the inequalities gives

$$-(|x| + |y|) \leq x + y \leq |x| + |y|,$$

which is equivalent to $|x + y| \leq |x| + |y|$.

Applying this triangle inequality to $x = (x - y) + y$ gives

$$|x| - |y| \leq |x - y|.$$

Interchanging x and y similarly gives $|y| - |x| \leq |x - y|$. Therefore

$$-|x - y| \leq |x| - |y| \leq |x - y|,$$

which is equivalent to the reverse triangle inequality. □

2.2. 03.09.2026.

2.2.1. More on absolute values.

Theorem 2.5. *We have*

$$|xy| = |x||y|$$

for all $x, y \in \mathbb{R}$.

Proof. We prove by splitting into four cases. Firstly, if $x, y \geq 0$, we have $xy \geq 0$, so $|xy| = xy$. But in this case also both $|x| = x$ and $|y| = y$, so $|xy| = xy = |x||y|$. Secondly, if $x, y < 0$, we have $xy > 0$, $|xy| = xy = (-x)(-y) = |x||y|$. Now suppose $x < 0$ and $y \geq 0$. Then $xy \leq 0$, and hence $|xy| = -xy = (-x)y = |x||y|$. The case $x \geq 0$, $y < 0$ follows from commutativity of multiplication of real numbers. □

Example 2.6. Let $x \in (1, 3)$ and $y \in (4, 8)$. Find an upper bound for

$$|x + y - 8|.$$

Kolmioepäyhtälön nojalla saadaan

$$\begin{aligned} |x + y - 8| &= |(x - 2) + (y - 6)| \\ &\leq |x - 2| + |y - 6| \\ &< 1 + 2 \\ &= 3. \end{aligned}$$

Thus we have found that 3 is an upper bound for $|x + y - 8|$.

Example 2.7. Let $x \in (2, 6)$ and $y \in (2, 8)$. Find an upper bound for

$$|xy - 20|.$$

We have $|x - 4| < 2$ and $|y - 5| < 3$. We have

$$\begin{aligned} |xy - 20| &= |xy - 4y + 4y - 20| \\ &\leq |xy - 4y| + |4y - 20| \\ &= |y||x - 4| + 4|y - 5| \\ &\leq 8 \cdot 2 + 4 \cdot 3 \\ &= 28. \end{aligned}$$

It follows that 28 is an upper bound for $|xy - 20|$.

Example 2.8. Find a positive real number m such that

$$|x^2 - 16| \leq m|x - 4|,$$

where $x \in (0, 10)$.

When $x = 4$, we get $0 \leq 0$, which holds with every $m \in \mathbb{R}$. Now suppose $x \neq 4$. We simplify the inequality and obtain

$$|x + 4| \leq m.$$

Since $x \in (0, 10)$, $|x + 4| \in (4, 14)$. We have thus obtained

$$|x + 4| \leq 14,$$

so we can choose $m = 14$.

2.2.2. Field and ordering properties of $\mathbb{R}$.

Definition 2.9. $(F, +, \cdot)$ is a field if and only if

- (1) The algebraic structure $(F, +)$ is an *abelian group*.
- (2) The algebraic structure $(F^*, \cdot)$ is an *abelian group*, where $F^* = F \setminus \{0_F\}$.
- (3) The operation $\cdot$ distributes over $+$.

Theorem 2.10. $\mathbb{R}$ is a field.

Proof. Consider the usual addition and multiplication on $\mathbb{R}$.

First, $(\mathbb{R}, +)$ is an abelian group. Addition is closed, associative, and commutative; 0 is an additive identity; and every $x \in \mathbb{R}$ has the additive inverse $-x$.

Next, $(\mathbb{R} \setminus \{0\}, \cdot)$ is an abelian group. Multiplication is associative and commutative, and 1 is its identity element. If $x \neq 0$, then $x^{-1} = 1/x$ is a nonzero real number satisfying

$$x \cdot x^{-1} = 1.$$

The product of two nonzero real numbers is nonzero, so multiplication is closed on $\mathbb{R} \setminus \{0\}$.

Finally, multiplication distributes over addition:

$$x(y + z) = xy + xz$$

for all $x, y, z \in \mathbb{R}$. Therefore $(\mathbb{R}, +, \cdot)$ is a field. □

Definition 2.11 (Ordered field). An *ordered field* is a field $(F, +, \cdot)$ equipped with a relation $<$ such that, for all $x, y, z \in F$,

- (1) exactly one of $x < y$, $x = y$, or $x > y$ holds;
- (2) if $x < y$ and $y < z$, then $x < z$;
- (3) if $x < y$, then $x + z < y + z$; and
- (4) if $0_F < x$ and $0_F < y$, then $0_F < xy$.

Here $x > y$ means $y < x$. The first two conditions state that $<$ is a strict total order, while the last two state that the order is compatible with the field operations.

Theorem 2.12. *The field $(\mathbb{R}, +, \cdot)$, equipped with the usual order $<$, is an ordered field.*

Proof. We have already shown that $(\mathbb{R}, +, \cdot)$ is a field. The usual strict order on $\mathbb{R}$ satisfies trichotomy and transitivity, so it satisfies the first two ordering axioms.

Suppose that $x < y$. Then $0 < y - x$, and for every $z \in \mathbb{R}$,

$$(y + z) - (x + z) = y - x > 0.$$

Hence $x + z < y + z$. Furthermore, if $0 < x$ and $0 < y$, then the usual sign rule for multiplication gives $0 < xy$. Thus the order is compatible with addition and multiplication, and $\mathbb{R}$ is an ordered field. $\square$

Theorem 2.13 (Uniqueness of additive identity). *The identity element 0_F of the abelian group $(F, +)$ is unique.*

Proof. Suppose that 0_F and $0'_F$ are both additive identity elements of $(F, +)$. Since 0_F is an additive identity,

$$0_F + 0'_F = 0'_F.$$

Since $0'_F$ is also an additive identity,

$$0_F + 0'_F = 0_F.$$

Therefore $0_F = 0'_F$. $\square$

Theorem 2.14 (Cancellation law). *Let $x, y, z \in \mathbb{R}$. If $x + y = x + z$, then $y = z$.*

Proof. Using the additive identity, additive inverse, associativity, and the assumption $x + y = x + z$, we obtain

$$\begin{aligned} y &= 0 + y \\ &= ((-x) + x) + y \\ &= (-x) + (x + y) \\ &= (-x) + (x + z) \\ &= ((-x) + x) + z \\ &= 0 + z \\ &= z. \end{aligned}$$

$\square$

2.3. 08.09.2026.

2.3.1. Sequences.

Definition 2.15. Let X and Y be sets. A function $f: X \rightarrow Y$ is a relation $f \subseteq X \times Y$ such that, for every $x \in X$, there exists exactly one $y \in Y$ with $(x, y) \in f$. This unique y is denoted by $f(x)$.

The set X is the *domain* of f , and Y is its *codomain*. The set

$$f(X) = \{f(x) : x \in X\}$$

is its *range* or *image*.

Definition 2.16. A sequence of real numbers is a function $x: \mathbb{N} \rightarrow \mathbb{R}$. For $n \in \mathbb{N}$, the value $x(n)$, usually denoted by x_n , is the n -th term of the sequence. We denote the sequence by $(x_n)_{n \in \mathbb{N}}$, or simply by (x_n) .

2.3.2. Limit of a sequence.

Definition 2.17. Let (x_n) be a sequence of real numbers. We say that (x_n) *converges* to $a \in \mathbb{R}$ if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that, for every $n \in \mathbb{N}$ with $n > N$,

$$|x_n - a| < \varepsilon.$$

In this case, we write $x_n \rightarrow a$ as $n \rightarrow \infty$, or $\lim_{n \rightarrow \infty} x_n = a$.

Let us look at some examples.

Example 2.18. Let (x_n) be defined by

$$x_n = \frac{2n}{n+1}.$$

We will show that

$$\lim_{n \rightarrow \infty} x_n = 2.$$

Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $N > \frac{2}{\varepsilon}$. Now for every $n > N$, we have

$$\begin{aligned} |x_n - 2| &= \left| \frac{2n}{n+1} - 2 \right| \\ &= \frac{2}{n+1} < \frac{2}{n} < \frac{2}{\frac{2}{\varepsilon}} = \varepsilon. \end{aligned}$$

Thus $x_n \rightarrow 2$ when $n \rightarrow \infty$.

Example 2.19. Let (x_n) be defined by

$$\begin{aligned} x_0 &= \cdots = x_7 = 42, \\ x_n &= 7 + \frac{1}{n}, \quad n \geq 8. \end{aligned}$$

We will show that

$$\lim_{n \rightarrow \infty} x_n = 7.$$

Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $N > \max \left\{ 8, \frac{1}{\varepsilon} \right\}$. Now for every $n > N$, we have

$$|x_n - 7| = \frac{1}{n} < \varepsilon.$$

Thus $x_n \rightarrow 7$ as $n \rightarrow \infty$.

Let us now tackle a simple divergence proof.

Example 2.20. Let us show that the sequence x_n defined by $x_n = n$ diverges. Assume that the sequence converged to some $a \in \mathbb{R}$. Taking $\varepsilon = 1$, there exists $N \in \mathbb{N}$ such that, for every $n > N$,

$$|x_n - a| < \varepsilon = 1.$$

Choose $n \in \mathbb{N}$ such that

$$n > \max \{N, |a| + 1\}.$$

By the triangle inequality,

$$n = |n| = |(n - a) + a| \leq |n - a| + |a|.$$

Subtracting $|a|$ from both sides and using $n > |a| + 1$, we obtain

$$|x_n - a| = |n - a| \geq n - |a| > 1,$$

contradicting the assumption $|x_n - a| < 1$. Therefore (x_n) diverges.

Example 2.21. Let (x_n) be defined by

$$\begin{aligned} x_0 &= \cdots = x_{42} = 1, \\ x_n &= \frac{n}{n-42}, \quad n \geq 43. \end{aligned}$$

We will show that

$$\lim_{n \rightarrow \infty} x_n = 1.$$

Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that

$$N > \max \left\{ 84, \frac{84}{\varepsilon} \right\}.$$

Now, for every $n > N$, we have $n > 84$, and hence $\frac{n}{2} - 42 > 0$. Therefore

$$\begin{aligned} |x_n - 1| &= \left| \frac{n}{n-42} - 1 \right| \\ &= \frac{42}{n-42} \\ &= \frac{42}{\frac{n}{2} + \frac{n}{2} - 42} \\ &< \frac{84}{n} < \varepsilon. \end{aligned}$$

Thus $x_n \rightarrow 1$ as $n \rightarrow \infty$.

2.4. 10.09.2026.

2.4.1. Uniqueness of limits.

Theorem 2.22 (Uniqueness of limits). *A convergent sequence of real numbers has a unique limit. That is, if $x_n \rightarrow a$ and $x_n \rightarrow b$ as $n \rightarrow \infty$, then $a = b$.*

Proof. Suppose for contradiction that $a \neq b$, and set

$$\varepsilon = \frac{|a - b|}{2} > 0.$$

Since $x_n \rightarrow a$ and $x_n \rightarrow b$, there exist $N_a, N_b \in \mathbb{N}$ such that

$$n > N_a \implies |x_n - a| < \varepsilon$$

and

$$n > N_b \implies |x_n - b| < \varepsilon.$$

Choose $n > \max \{N_a, N_b\}$. By the triangle inequality,

$$|a - b| \leq |a - x_n| + |x_n - b| < 2\varepsilon = |a - b|,$$

which is impossible. Therefore $a = b$. □

Theorem 2.23 (Convergent sequences are bounded). *Every convergent sequence of real numbers is bounded.*

Proof. Suppose that $x_n \rightarrow a$. Taking $\varepsilon = 1$, there exists $N \in \mathbb{N}$ such that

$$n > N \implies |x_n - a| < 1.$$

By the triangle inequality, for every $n > N$,

$$|x_n| \leq |x_n - a| + |a| < 1 + |a|.$$

Let

$$M = \max \{|x_0|, |x_1|, \dots, |x_N|, 1 + |a|\}.$$

Then $|x_n| \leq M$ for every $n \in \mathbb{N}$, so (x_n) is bounded. □

2.4.2. Algebra of limits.

Theorem 2.24 (Algebraic limit laws). *Suppose that $x_n \rightarrow a$ and $y_n \rightarrow b$, and let $c \in \mathbb{R}$. Then*

- (1) $x_n \pm y_n \rightarrow a \pm b$.
- (2) $cx_n \rightarrow ca$.
- (3) $x_n y_n \rightarrow ab$.
- (4) If $b \neq 0$, then $y_n \neq 0$ for all sufficiently large n , and

$$\frac{x_n}{y_n} \rightarrow \frac{a}{b}.$$

Proof. (1) Let $\varepsilon > 0$. Since $x_n \rightarrow a$ and $y_n \rightarrow b$, there exist $N_x, N_y \in \mathbb{N}$ such that

$$n > N_x \implies |x_n - a| < \frac{\varepsilon}{2}$$

and

$$n > N_y \implies |y_n - b| < \frac{\varepsilon}{2}.$$

Thus, for $n > \max\{N_x, N_y\}$,

$$|(x_n + y_n) - (a + b)| \leq |x_n - a| + |y_n - b| < \varepsilon.$$

Hence $x_n + y_n \rightarrow a + b$. Replacing y_n and b by $-y_n$ and $-b$ gives $x_n - y_n \rightarrow a - b$.

- (2) If $c = 0$, then $cx_n = ca = 0$ for every n . If $c \neq 0$, choose $N \in \mathbb{N}$ such that

$$n > N \implies |x_n - a| < \frac{\varepsilon}{|c|}.$$

Then, for every $n > N$,

$$|cx_n - ca| = |c||x_n - a| < \varepsilon.$$

Therefore $cx_n \rightarrow ca$.

- (3) Since (y_n) converges, it is bounded. Choose $M > 0$ such that $|y_n| \leq M$ for every n . There exist $N_1, N_2 \in \mathbb{N}$ such that

$$n > N_1 \implies |x_n - a| < \frac{\varepsilon}{2M}$$

and

$$n > N_2 \implies |y_n - b| < \frac{\varepsilon}{2(|a| + 1)}.$$

For $n > \max\{N_1, N_2\}$,

$$\begin{aligned} |x_n y_n - ab| &\leq |x_n y_n - a y_n| + |a y_n - ab| \\ &= |y_n| |x_n - a| + |a| |y_n - b| \\ &< M \frac{\varepsilon}{2M} + |a| \frac{\varepsilon}{2(|a| + 1)} \\ &< \varepsilon. \end{aligned}$$

Therefore $x_n y_n \rightarrow ab$.

- (4) Suppose that $b \neq 0$. Since $y_n \rightarrow b$, there exists $N_0 \in \mathbb{N}$ such that

$$n > N_0 \implies |y_n - b| < \frac{|b|}{2}.$$

The reverse triangle inequality then gives

$$|y_n| \geq |b| - |y_n - b| > \frac{|b|}{2} > 0$$

for every $n > N_0$. In particular, $y_n \neq 0$ eventually. Choose $N_1 \in \mathbb{N}$ such that

$$n > N_1 \implies |y_n - b| < \frac{\varepsilon|b|^2}{2}.$$

For $n > \max\{N_0, N_1\}$,

$$\left| \frac{1}{y_n} - \frac{1}{b} \right| = \frac{|y_n - b|}{|y_n||b|} < \frac{2}{|b|^2} |y_n - b| < \varepsilon.$$

Hence $1/y_n \rightarrow 1/b$; the finitely many terms with $n \leq N_0$ may be defined arbitrarily because they do not affect the limit. Applying the product law to (x_n) and $(1/y_n)$ yields

$$\frac{x_n}{y_n} \rightarrow \frac{a}{b}.$$

□

2.4.3. Limits and order.

Theorem 2.25. *Suppose that $x_n \rightarrow x$. Then*

- (1) *if $x_n \leq M$ for every $n \in \mathbb{N}$, then $x \leq M$;*
- (2) *if $x_n \geq m$ for every $n \in \mathbb{N}$, then $x \geq m$.*

Proof. (1) Suppose for contradiction that $x > M$, and set

$$\varepsilon = \frac{x - M}{2} > 0.$$

Since $x_n \rightarrow x$, there exists $N \in \mathbb{N}$ such that, for every $n > N$,

$$|x_n - x| < \varepsilon.$$

It follows that

$$x_n > x - \varepsilon = x - \frac{x - M}{2} = \frac{x + M}{2} > M,$$

contradicting $x_n \leq M$. Therefore $x \leq M$.

- (2) Suppose for contradiction that $x < m$, and set

$$\varepsilon = \frac{m - x}{2} > 0.$$

Since $x_n \rightarrow x$, there exists $N \in \mathbb{N}$ such that, for every $n > N$,

$$|x_n - x| < \varepsilon.$$

It follows that

$$x_n < x + \varepsilon = x + \frac{m - x}{2} = \frac{x + m}{2} < m,$$

contradicting $x_n \geq m$. Therefore $x \geq m$.

□

Theorem 2.26 (Squeeze theorem). *Suppose that*

$$x_n \leq y_n \leq z_n$$

for every $n \in \mathbb{N}$. If $x_n \rightarrow a$ and $z_n \rightarrow a$, then $y_n \rightarrow a$.

Proof. Let $\varepsilon > 0$. Since $x_n \rightarrow a$, there exists $N_x \in \mathbb{N}$ such that

$$n > N_x \implies a - \varepsilon < x_n < a + \varepsilon.$$

Similarly, since $z_n \rightarrow a$, there exists $N_z \in \mathbb{N}$ such that

$$n > N_z \implies a - \varepsilon < z_n < a + \varepsilon.$$

Therefore, for every $n > \max \{N_x, N_z\}$,

$$a - \varepsilon < x_n \leq y_n \leq z_n < a + \varepsilon.$$

Hence $|y_n - a| < \varepsilon$, so $y_n \rightarrow a$. $\square$

The inequalities in the squeeze theorem need only hold for all sufficiently large n , since changing finitely many terms does not affect convergence.

Example 2.27. Consider the oscillating sequence

$$x_n = \frac{(-1)^n}{n}, \quad n \geq 1.$$

Its sign alternates, but

$$-\frac{1}{n} \leq \frac{(-1)^n}{n} \leq \frac{1}{n}.$$

Both bounding sequences converge to 0. Indeed, given $\varepsilon > 0$, choosing $N > 1/\varepsilon$ gives $1/n < \varepsilon$ whenever $n > N$. The squeeze theorem therefore gives

$$\lim_{n \rightarrow \infty} \frac{(-1)^n}{n} = 0.$$

Notice that the squeeze theorem lets us prove convergence without trying to handle even and odd indices separately.

Example 2.28. Consider

$$x_n = \sqrt{n^2 + n} - n, \quad n \geq 1.$$

Rationalising the expression gives

$$x_n = \frac{n}{\sqrt{n^2 + n} + n}.$$

Since

$$n < \sqrt{n^2 + n} < n + 1,$$

we obtain

$$\frac{n}{2n + 1} < x_n < \frac{1}{2}.$$

By the algebraic limit laws,

$$\lim_{n \rightarrow \infty} \frac{n}{2n + 1} = \lim_{n \rightarrow \infty} \frac{1}{2 + 1/n} = \frac{1}{2}.$$

The upper bound is the constant sequence $1/2$, so both bounds converge to $1/2$. Hence the squeeze theorem yields

$$\lim_{n \rightarrow \infty} (\sqrt{n^2 + n} - n) = \frac{1}{2}.$$

2.4.4. Subsequences.

Definition 2.29. Let (x_n) be a sequence, and let (n_k) be a strictly increasing sequence of natural numbers:

$$n_0 < n_1 < n_2 < \dots$$

The sequence (x_{n_k}) , obtained by selecting the terms of (x_n) whose indices are $n_0, n_1, n_2, \dots$, is called a *subsequence* of (x_n) .

Theorem 2.30. If $x_n \rightarrow x$, then every subsequence of (x_n) converges to x .

Proof. Let (x_{n_k}) be a subsequence of (x_n) , and let $\varepsilon > 0$. Since $x_n \rightarrow x$, there exists $N \in \mathbb{N}$ such that

$$n > N \implies |x_n - x| < \varepsilon.$$

Because (n_k) is strictly increasing, $n_k \geq k$ for every $k \in \mathbb{N}$. Thus, whenever $k > N$, we also have $n_k > N$, and hence

$$|x_{n_k} - x| < \varepsilon.$$

Therefore $x_{n_k} \rightarrow x$. Since the subsequence was arbitrary, every subsequence of (x_n) converges to the same limit x . $\square$

2.5. 15.09.2026.

2.5.1. Upper and lower bounds.

Definition 2.31. Let $\emptyset \neq E \subseteq \mathbb{R}$. A number $s \in \mathbb{R}$ is an *upper bound* of E if

$$x \leq s \quad \text{for every } x \in E.$$

If such a number exists, E is *bounded above*.

Definition 2.32. Let $\emptyset \neq E \subseteq \mathbb{R}$. A number $p \in \mathbb{R}$ is a *lower bound* of E if

$$p \leq x \quad \text{for every } x \in E.$$

If such a number exists, E is *bounded below*.

Definition 2.33. Let $\emptyset \neq E \subseteq \mathbb{R}$ be bounded above. An upper bound s of E is the *least upper bound*, or *supremum*, of E if $s \leq u$ for every upper bound u of E . We write

$$s = \sup E.$$

Definition 2.34. Let $\emptyset \neq E \subseteq \mathbb{R}$ be bounded below. A lower bound p of E is the *greatest lower bound*, or *infimum*, of E if $p \geq \ell$ for every lower bound ℓ of E . We write

$$p = \inf E.$$

If E is not bounded above, we write $\sup E = +\infty$. Similarly, if E is not bounded below, we write $\inf E = -\infty$.

2.5.2. The completeness axiom.

Completeness axiom. Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.

Corollary 2.35. Every nonempty subset $E \subseteq \mathbb{R}$ that is bounded below has an infimum in $\mathbb{R}$.

Proof. Let c be a lower bound of E , and define

$$F = \{-x : x \in E\}.$$

Since $c \leq x$ for every $x \in E$, we have $-x \leq -c$. Thus F is bounded above. It is also nonempty, so the completeness axiom gives $\sup F \in \mathbb{R}$. For every $x \in E$,

$$-x \leq \sup F \leq -c, \quad \text{hence} \quad x \geq -\sup F \geq c.$$

Therefore $-\sup F$ is a lower bound of E . Since the argument applies to every lower bound c of E , it is the greatest lower bound. Consequently,

$$\inf E = -\sup F \in \mathbb{R}.$$

$\square$

2.5.3. Maximum and properties of the supremum.

Definition 2.36. A number s is the *greatest element*, or *maximum*, of a set E if $s \in E$ and $x \leq s$ for every $x \in E$. We write $s = \max E$.

Theorem 2.37. Let $\emptyset \neq E \subseteq \mathbb{R}$ be bounded above. If $s = \max E$, then $s = \sup E$.

Proof. Since $x \leq s$ for every $x \in E$, the number s is an upper bound of E . Hence $\sup E \leq s$. On the other hand, $s \in E$, so $s \leq \sup E$. Thus $s = \sup E$. $\square$

Theorem 2.38. The supremum of a nonempty subset of $\mathbb{R}$ that is bounded above is unique.

Proof. Suppose that s and s' are both least upper bounds of E . Since s' is an upper bound and s is a least upper bound, $s \leq s'$. Similarly, since s is an upper bound and s' is a least upper bound, $s' \leq s$. Therefore $s = s'$. $\square$

Theorem 2.39. Let $\emptyset \neq E \subseteq \mathbb{R}$ be bounded above. For every $\varepsilon > 0$, there exists $x \in E$ such that

$$\sup E - \varepsilon < x \leq \sup E.$$

Proof. Since $\sup E$ is an upper bound, $x \leq \sup E$ for every $x \in E$. Suppose, for a contradiction, that for some $\varepsilon > 0$,

$$x \leq \sup E - \varepsilon \quad \text{for every } x \in E.$$

Then $\sup E - \varepsilon$ is an upper bound of E , but $\sup E - \varepsilon < \sup E$. This contradicts the fact that $\sup E$ is the least upper bound. Hence the claimed element x exists. $\square$

2.5.4. Monotone sequences.

Definition 2.40. A real sequence (x_n) is

- *increasing* (nondecreasing) if $x_n \leq x_{n+1}$ for every $n \in \mathbb{N}$;
- *decreasing* (nonincreasing) if $x_{n+1} \leq x_n$ for every $n \in \mathbb{N}$;
- *strictly increasing* if $x_n < x_{n+1}$ for every $n \in \mathbb{N}$;
- *strictly decreasing* if $x_{n+1} < x_n$ for every $n \in \mathbb{N}$.

A sequence is *monotone* if it is increasing or decreasing, and *strictly monotone* if it is strictly increasing or strictly decreasing.

2.6. 17.09.2026.

2.6.1. Monotone convergence.

Theorem 2.41 (Monotone convergence theorem). Let (x_n) be an increasing sequence of real numbers. If there is $M \in \mathbb{R}$ such that $x_n \leq M$ for every $n \in \mathbb{N}$, then (x_n) converges, and

$$\lim_{n \rightarrow \infty} x_n = \sup \{x_n : n \in \mathbb{N}\} \leq M.$$

Proof. The set $E = \{x_n : n \in \mathbb{N}\}$ is nonempty and bounded above by M . By completeness, $s = \sup E$ exists in $\mathbb{R}$, and $s \leq M$. Let $\varepsilon > 0$. By the approximation property of the supremum, there is an index $N \in \mathbb{N}$ such that

$$s - \varepsilon < x_N \leq s.$$

Since (x_n) is increasing, for every $n > N$,

$$s - \varepsilon < x_N \leq x_n \leq s.$$

Hence $|x_n - s| = s - x_n < \varepsilon$, proving $x_n \rightarrow s$. $\square$

Corollary 2.42 (Decreasing case). *Let (x_n) be a decreasing sequence of real numbers. If there is $m \in \mathbb{R}$ such that $m \leq x_n$ for every $n \in \mathbb{N}$, then (x_n) converges, and*

$$\lim_{n \rightarrow \infty} x_n = \inf \{x_n : n \in \mathbb{N}\} \geq m.$$

Consequently, every bounded monotone real sequence converges.

Proof. The sequence $(-x_n)$ is increasing and bounded above by $-m$. By theorem 2.41,

$$-x_n \longrightarrow \sup \{-x_n : n \in \mathbb{N}\}.$$

Multiplying by -1 and using the relation between infima and suprema gives

$$x_n \longrightarrow -\sup \{-x_n : n \in \mathbb{N}\} = \inf \{x_n : n \in \mathbb{N}\} \geq m.$$

A bounded monotone sequence satisfies the hypotheses of either the increasing or the decreasing case, so it converges. $\square$

2.6.2. *Examples of decreasing sequences.* In the following two examples the indices start at $n = 1$.

Example 2.43. Let

$$x_1 = 3, \quad x_n = \frac{n}{n-1} \quad (n \geq 2).$$

Show that (x_n) is strictly decreasing and find its limit.

Solution. First, $x_2 = 2 < 3 = x_1$. For $n \geq 2$, all terms are positive and

$$\frac{x_{n+1}}{x_n} = \frac{(n+1)(n-1)}{n^2} = 1 - \frac{1}{n^2} < 1.$$

Thus $x_{n+1} < x_n$ for every $n \geq 1$. Also $x_1 > 1$, and

$$x_n = 1 + \frac{1}{n-1} > 1 \quad (n \geq 2).$$

Hence the sequence is bounded below and converges by corollary 2.42. We identify its limit in two ways.

Method 1: the squeeze theorem. For $n \geq 2$, we have $n-1 \geq n/2$, so

$$1 < x_n = 1 + \frac{1}{n-1} \leq 1 + \frac{2}{n}.$$

Both bounding sequences tend to 1, so $x_n \rightarrow 1$.

Method 2: the infimum. The number 1 is a lower bound of $E = \{x_n : n \geq 1\}$. If $a > 1$, choose an integer $n > 1 + 1/(a-1)$. Then

$$x_n < a \iff \frac{1}{n-1} < a-1 \iff n > 1 + \frac{1}{a-1}.$$

Thus no $a > 1$ is a lower bound of E . Therefore $\inf E = 1$, and the decreasing case of the monotone convergence theorem yields $\lim_{n \rightarrow \infty} x_n = 1$. $\square$

Example 2.44. Let

$$x_n = \frac{(n+1)^2}{n^2+1}, \quad n \geq 1.$$

Show that (x_n) is decreasing and determine its limit.

Solution. Since every term is positive, we can compare successive terms using their ratio:

$$\begin{aligned} \frac{x_{n+1}}{x_n} &= \frac{(n+2)^2(n^2+1)}{((n+1)^2+1)(n+1)^2} \\ &= \frac{n^4+4n^3+5n^2+4n+4}{n^4+4n^3+7n^2+6n+2}. \end{aligned}$$

The denominator exceeds the numerator by

$$2n^2 + 2n - 2 = 2(n^2 + n - 1) > 0 \quad (n \geq 1).$$

Therefore $x_{n+1}/x_n < 1$, so the sequence is strictly decreasing. Moreover,

$$x_n = 1 + \frac{2n}{n^2 + 1} > 1,$$

so it is bounded below and converges to $\inf \{x_n : n \geq 1\}$.

We now show that this infimum is 1. It is already a lower bound. Given $a > 1$, choose an integer $n > \max \{1, 2/(a-1)\}$. Then

$$x_n = 1 + \frac{2n}{n^2 + 1} < 1 + \frac{2}{n} < a.$$

Hence a is not a lower bound, and consequently

$$\lim_{n \rightarrow \infty} x_n = \inf \{x_n : n \geq 1\} = 1.$$

Alternatively, $1 < x_n < 1 + 2/n$ gives the same limit directly by the squeeze theorem. $\square$

Remark 2.45. The restriction $n \geq 1$ matters for monotonicity in the second example. If we also define x_0 by the same formula, then $x_0 = 1 < x_1 = 2$, so the full sequence is not decreasing. Its tail starting at $n = 1$ is strictly decreasing, and its limit is still 1.

2.6.3. The Bolzano–Weierstrass theorem.

Theorem 2.46 (Bolzano–Weierstrass). *Every bounded sequence of real numbers has a convergent subsequence.*

Proof. Let (x_n) be bounded. Choose $m, M \in \mathbb{R}$ such that $m \leq x_n \leq M$ for every $n \in \mathbb{N}$, and define the *tail suprema*

$$a_k = \sup \{x_n : n \geq k\}, \quad k \in \mathbb{N}.$$

Each tail is nonempty and bounded, so completeness ensures that a_k exists and $m \leq a_k \leq M$. Since

$$\{x_n : n \geq k+1\} \subseteq \{x_n : n \geq k\},$$

the number a_k is an upper bound of the smaller tail as well. Therefore $a_{k+1} \leq a_k$. The sequence (a_k) is decreasing and bounded below, so corollary 2.42 gives a limit

$$a = \lim_{k \rightarrow \infty} a_k \in \mathbb{R}.$$

We construct a subsequence $(x_{n_j})_{j \geq 1}$ recursively. By the approximation property of the supremum, choose $n_1 \geq 1$ such that

$$a_1 - 1 < x_{n_1} \leq a_1.$$

Once n_j has been chosen, apply the same property to the tail starting at $n_j + 1$. There is an index $n_{j+1} \geq n_j + 1$ such that

$$a_{n_j+1} - \frac{1}{j+1} < x_{n_{j+1}} \leq a_{n_j+1}.$$

Thus the indices are strictly increasing, as required for a subsequence.

To verify convergence, observe that $a_{n_j+1} \leq a_{n_j+1}$ and $x_{n_{j+1}} \leq a_{n_j+1}$. Consequently, for every $j \geq 1$,

$$a_{n_j} - \frac{1}{j} < x_{n_j} \leq a_{n_j}.$$

For $j = 1$, the lower inequality follows from $a_{n_1} \leq a_1$; for later indices it follows from the recursive choice above. Since (a_{n_j}) is a subsequence of the convergent sequence (a_k) , it tends to a . Also $1/j \rightarrow 0$, so both bounding sequences tend to a . The squeeze theorem now yields $x_{n_j} \rightarrow a$, proving the claim. $\square$

2.7. 22.09.2026.

2.7.1. Bernoulli's inequality.

Theorem 2.47 (Bernoulli's inequality). *For every $x \in \mathbb{R}$ with $x > -1$ and every $n \in \mathbb{N}$,*

$$(1+x)^n \geq 1+nx.$$

Proof. Fix $x > -1$. The cases $n = 0$ and $n = 1$ are equalities. Suppose that $(1+x)^n \geq 1+nx$ for some $n \geq 1$. Since $1+x > 0$, multiplication preserves the inequality, and

$$\begin{aligned} (1+x)^{n+1} &= (1+x)(1+x)^n \\ &\geq (1+x)(1+nx) \\ &= 1 + (n+1)x + nx^2 \\ &\geq 1 + (n+1)x. \end{aligned}$$

The result follows by induction. If $x \neq 0$, then $nx^2 > 0$ in the induction step, so the inequality is strict for $n \geq 2$. $\square$

2.7.2. A sequence defining the number e . Consider the sequence

$$x_n = \left(1 + \frac{1}{n}\right)^n, \quad n \geq 1.$$

Lemma 2.48. *The sequence (x_n) is strictly increasing.*

Proof. All terms are positive. For $n \geq 1$,

$$\begin{aligned} \frac{x_{n+1}}{x_n} &= \frac{\left(1 + \frac{1}{n+1}\right)^{n+1}}{\left(1 + \frac{1}{n}\right)^n} \\ &= \left(\frac{1 + \frac{1}{n+1}}{1 + \frac{1}{n}}\right)^{n+1} \left(1 + \frac{1}{n}\right) \\ &= \left(\frac{n(n+2)}{(n+1)^2}\right)^{n+1} \frac{n+1}{n} \\ &= \left(1 - \frac{1}{(n+1)^2}\right)^{n+1} \frac{n+1}{n} \\ &> \left(1 - \frac{n+1}{(n+1)^2}\right) \frac{n+1}{n} = 1. \end{aligned}$$

The strict inequality follows from theorem 2.47, since $-1 < -1/(n+1)^2 < 0$ and $n+1 \geq 2$. Hence $x_{n+1} > x_n$. $\square$

Lemma 2.49. *The sequence (x_n) is bounded above.*

Proof. We use the auxiliary sequence

$$y_n = \left(1 - \frac{1}{n}\right)^n, \quad n \geq 2.$$

This sequence is positive and strictly increasing. Indeed, by theorem 2.47, for $n \geq 2$,

$$\begin{aligned}\frac{y_{n+1}}{y_n} &= \left(\frac{n^2}{n^2-1}\right)^n \frac{n}{n+1} \\ &= \left(1 + \frac{1}{n^2-1}\right)^n \frac{n}{n+1} \\ &\geq \left(1 + \frac{n}{n^2-1}\right) \frac{n}{n+1} \\ &= 1 + \frac{1}{(n+1)(n^2-1)} > 1.\end{aligned}$$

Consequently, $(1/y_n)$ is decreasing. Also, for $n \geq 2$,

$$\left(1 + \frac{1}{n}\right) \left(1 - \frac{1}{n}\right) = 1 - \frac{1}{n^2} < 1.$$

Dividing by $1 - 1/n > 0$ and raising to the power n gives

$$x_n < \left(1 - \frac{1}{n}\right)^{-n} = \frac{1}{y_n} \leq \frac{1}{y_2} = \left(1 - \frac{1}{2}\right)^{-2} = 4.$$

Finally, $x_1 = 2 < 4$, so the bound holds for every $n \geq 1$. $\square$

By theorem 2.41, the increasing, bounded sequence (x_n) converges to a real number.

Definition 2.50. The limit

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$$

is called *Euler's number* (*Neperin luku* in Finnish).

2.8. 24.09.2026.

2.8.1. Sequences tending to infinity.

Definition 2.51. A real sequence (x_n) *tends to positive infinity* if, for every $M \in \mathbb{R}$, there exists $N \in \mathbb{N}$ such that

$$x_n > M \quad \text{whenever } n > N.$$

We write $x_n \rightarrow +\infty$, or $\lim_{n \rightarrow \infty} x_n = +\infty$.

Definition 2.52. A real sequence (x_n) *tends to negative infinity* if, for every $M \in \mathbb{R}$, there exists $N \in \mathbb{N}$ such that

$$x_n < M \quad \text{whenever } n > N.$$

We write $x_n \rightarrow -\infty$, or $\lim_{n \rightarrow \infty} x_n = -\infty$.

Remark 2.53. Neither definition requires the sequence to be monotone. The symbols $+\infty$ and $-\infty$ are not real numbers, so these statements do not mean convergence to a real limit.

Theorem 2.54. *If $x_n \rightarrow +\infty$ or $x_n \rightarrow -\infty$, then (x_n) diverges as a sequence of real numbers.*

Proof. Suppose first that $x_n \rightarrow +\infty$, and assume for a contradiction that $x_n \rightarrow x \in \mathbb{R}$. Taking $M = x + 1$ in the definition of an infinite limit gives $N_1 \in \mathbb{N}$ such that $x_n > x + 1$ whenever $n > N_1$. Convergence to x , with $\varepsilon = 1$, gives $N_2 \in \mathbb{N}$ such that $|x_n - x| < 1$ whenever $n > N_2$. For $n > \max\{N_1, N_2\}$, we would have both $|x_n - x| > 1$ and $|x_n - x| < 1$, a contradiction.

If $x_n \rightarrow -\infty$, the same argument uses $M = x - 1$. Eventually $x_n < x - 1$, hence $|x_n - x| > 1$, again contradicting convergence to x . $\square$

2.8.2. A necessary condition for convergence.

Theorem 2.55. *Let (x_n) be a convergent real sequence. For every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that*

$$|x_n - x_{n+k}| < \varepsilon \quad \text{for all } n > N \text{ and } k \in \mathbb{N}.$$

Proof. Write $x = \lim_{n \rightarrow \infty} x_n$, and let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $|x_m - x| < \varepsilon/2$ for every $m > N$. If $n > N$ and $k \in \mathbb{N}$, then $n+k > N$, so the triangle inequality gives

$$|x_n - x_{n+k}| \leq |x_n - x| + |x_{n+k} - x| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

□

Definition 2.56. A real sequence satisfying the condition in theorem 2.55 is called a *Cauchy sequence*. Equivalently, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|x_p - x_q| < \varepsilon \quad \text{whenever } p, q > N.$$

Indeed, take $n = \min\{p, q\}$ and $k = |p - q|$ to obtain this form from the preceding one. Conversely, set $p = n$ and $q = n + k$.

2.8.3. Nested closed intervals.

Lemma 2.57 (Nested interval lemma). *Let $[a_n, b_n]$, $n \geq 1$, be nonempty closed intervals such that*

$$[a_{n+1}, b_{n+1}] \subseteq [a_n, b_n] \quad (n \geq 1), \quad \lim_{n \rightarrow \infty} (b_n - a_n) = 0.$$

Then $\bigcap_{n=1}^{\infty} [a_n, b_n]$ contains exactly one real number.

Proof. The nesting gives

$$a_n \leq a_{n+1} \leq b_{n+1} \leq b_n.$$

Thus (a_n) is increasing and (b_n) is decreasing. Both sequences lie in $[a_1, b_1]$, so the monotone convergence theorem gives real limits

$$a = \lim_{n \rightarrow \infty} a_n, \quad b = \lim_{n \rightarrow \infty} b_n.$$

Since $a_n \leq b_n$, preservation of inequalities under limits yields $a \leq b$. Monotonicity also gives $a_n \leq a \leq b \leq b_n$ for every n , hence

$$0 \leq b - a \leq b_n - a_n.$$

The right-hand side tends to zero, so $a = b$. In particular, $a \in [a_n, b_n]$ for every n , proving existence of a point in the intersection.

For uniqueness, let $y \neq a$. Since $|y - a| > 0$ and $b_n - a_n \rightarrow 0$, choose n with $b_n - a_n < |y - a|$. If y also belonged to $[a_n, b_n]$, then $a, y \in [a_n, b_n]$ would imply $|y - a| \leq b_n - a_n$, a contradiction. Therefore y does not belong to the intersection, and

$$\bigcap_{n=1}^{\infty} [a_n, b_n] = \{a\}.$$

□

2.8.4. Cauchy's convergence criterion.

Theorem 2.58 (Cauchy's convergence criterion). *A real sequence (x_n) converges if and only if it is a Cauchy sequence: for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that*

$$|x_n - x_{n+k}| < \varepsilon \quad \text{for all } n > N \text{ and } k \in \mathbb{N}.$$

Proof. Necessity was proved in theorem 2.55. Conversely, suppose that (x_n) satisfies the Cauchy condition.

Step 1: boundedness. Apply the condition with $\varepsilon = 1$, and choose a corresponding $N_0 \in \mathbb{N}$. For every $m \geq N_0 + 1$, take $n = N_0 + 1$ and $k = m - (N_0 + 1)$. Then

$$|x_m - x_{N_0+1}| < 1, \quad |x_m| < |x_{N_0+1}| + 1.$$

The finitely many earlier terms are bounded as well. In particular,

$$C = \max\{|x_0|, \dots, |x_{N_0}|, |x_{N_0+1}| + 1\}$$

satisfies $|x_m| \leq C$ for every $m \in \mathbb{N}$.

Step 2: nested intervals from the tails. Define, for $n \geq 1$,

$$p_n = \inf \{x_m : m \geq n\}, \quad s_n = \sup \{x_m : m \geq n\}.$$

Each tail is nonempty and bounded, so its infimum and supremum exist in $\mathbb{R}$. As the tails shrink, their infima increase and their suprema decrease:

$$p_n \leq p_{n+1} \leq s_{n+1} \leq s_n.$$

Consequently, the nonempty closed intervals $[p_n, s_n]$ are nested.

Step 3: the interval lengths tend to zero. Let $\varepsilon > 0$. By the Cauchy condition, choose N so that $|x_r - x_t| < \varepsilon/3$ whenever $r, t > N$. Fix $n > N$. By the approximation properties of the supremum and infimum, there are indices $r, t \geq n$ such that

$$0 \leq s_n - x_r < \frac{\varepsilon}{3}, \quad 0 \leq x_t - p_n < \frac{\varepsilon}{3}.$$

Since $r, t > N$, we obtain

$$\begin{aligned} 0 \leq s_n - p_n &\leq |s_n - x_r| + |x_r - x_t| + |x_t - p_n| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

Hence $s_n - p_n \rightarrow 0$.

Step 4: convergence to the common point. By lemma 2.57, there is exactly one $x \in \mathbb{R}$ such that

$$\bigcap_{n=1}^{\infty} [p_n, s_n] = \{x\}.$$

Given $\varepsilon > 0$, choose n with $s_n - p_n < \varepsilon$. For every $m \geq n$, both x and x_m belong to $[p_n, s_n]$, so

$$|x_m - x| \leq s_n - p_n < \varepsilon.$$

This proves $x_m \rightarrow x$, completing the sufficiency direction. $\square$

2.9. 29.09.2026.

2.9.1. Limits of functions.

Definition 2.59. Let $f : A \rightarrow \mathbb{R}$, where $A \subseteq \mathbb{R}$, and let $x_0 \in \mathbb{R}$. Suppose that there is $r > 0$ such that

$$(x_0 - r, x_0 + r) \setminus \{x_0\} \subseteq A.$$

The function f has *limit* $m \in \mathbb{R}$ at x_0 if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|f(x) - m| < \varepsilon \quad \text{whenever } x \in A \text{ and } 0 < |x - x_0| < \delta.$$

We write $\lim_{x \rightarrow x_0} f(x) = m$.

Example 2.60. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 2x + 1$. Then $\lim_{x \rightarrow 1} f(x) = 3$.

Proof. Given $\varepsilon > 0$, choose $\delta = \varepsilon/2$. If $0 < |x - 1| < \delta$, then

$$|f(x) - 3| = |2x - 2| = 2|x - 1| < 2\delta = \varepsilon.$$

$\square$

Example 2.61. Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} 2x + 1, & x \neq 1, \\ 42, & x = 1. \end{cases}$$

Again, $\lim_{x \rightarrow 1} f(x) = 3$, with exactly the same proof: the condition $0 < |x - 1| < \delta$ excludes $x = 1$. Thus the value of a function at the point does not affect its limit.

2.9.2. Linear estimates and Lipschitz continuity.

Example 2.62. Suppose that, for some $C > 0$, $r > 0$, and $L \in \mathbb{R}$,

$$|f(x) - L| \leq C|x - x_0| \quad \text{whenever } 0 < |x - x_0| < r.$$

Then $\lim_{x \rightarrow x_0} f(x) = L$. Indeed, given $\varepsilon > 0$, choose $\delta = \min\{r, \varepsilon/C\}$. For $0 < |x - x_0| < \delta$, we have

$$|f(x) - L| \leq C|x - x_0| < C\delta \leq \varepsilon.$$

If the estimate holds throughout the domain, we may simply take $\delta = \varepsilon/C$.

Definition 2.63. A function $f : A \rightarrow \mathbb{R}$ is *Lipschitz continuous* if there exists a constant $L \geq 0$ such that

$$|f(x) - f(s)| \leq L|x - s| \quad \text{for all } x, s \in A.$$

Example 2.64. The function $f : [0, 1] \rightarrow \mathbb{R}$, $f(x) = \sqrt{x}$, is not Lipschitz continuous. For every integer $n \geq 1$,

$$\frac{|f(1/n) - f(0)|}{|1/n - 0|} = \frac{\sqrt{1/n}}{1/n} = \sqrt{n} \rightarrow +\infty.$$

A Lipschitz constant would have to bound all these quotients, which is impossible.

2.9.3. Choosing a local bound.

Example 2.65. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$. Then $\lim_{x \rightarrow 2} f(x) = 4$.

Proof. Factor the difference:

$$|f(x) - 4| = |x + 2||x - 2|.$$

If $|x - 2| < 1$, the triangle inequality gives $|x + 2| \leq |x - 2| + 4 < 5$. Given $\varepsilon > 0$, choose $\delta = \min\{1, \varepsilon/5\}$. Whenever $0 < |x - 2| < \delta$, it follows that

$$|f(x) - 4| = |x + 2||x - 2| < 5\delta \leq \varepsilon.$$

□

Example 2.66 (Positivity near a point with positive limit). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and suppose that $\lim_{x \rightarrow x_0} f(x) = m > 0$. Then there exists $r > 0$ such that

$$f(x) > 0 \quad \text{for every } x \in (x_0 - r, x_0 + r) \setminus \{x_0\}.$$

Proof. Since $m > 0$, take $\varepsilon = m$ in the definition of the limit. There exists $\delta > 0$ such that $|f(x) - m| < m$ whenever $0 < |x - x_0| < \delta$. Hence

$$-m < f(x) - m < m, \quad \text{so} \quad 0 < f(x) < 2m.$$

Taking $r = \delta$ proves the claim.

□

2.9.4. *Uniqueness of the limit.*

Theorem 2.67. *The limit of a function at a point, if it exists, is unique.*

Proof. Let $f : A \rightarrow \mathbb{R}$, and suppose that $(x_0 - r, x_0 + r) \setminus \{x_0\} \subseteq A$ for some $r > 0$. Assume, for a contradiction, that

$$\lim_{x \rightarrow x_0} f(x) = a, \quad \lim_{x \rightarrow x_0} f(x) = b, \quad a \neq b.$$

Set $\varepsilon = |a - b|/2 > 0$. There exist $\delta_1, \delta_2 > 0$ such that

$$0 < |x - x_0| < \delta_1 \implies |f(x) - a| < \varepsilon,$$

$$0 < |x - x_0| < \delta_2 \implies |f(x) - b| < \varepsilon.$$

Let $\delta = \min\{r, \delta_1, \delta_2\} > 0$, and choose $x = x_0 + \delta/2$. Then $x \in A$ and $0 < |x - x_0| < \delta$, so both estimates hold. The triangle inequality now gives

$$|a - b| \leq |f(x) - a| + |f(x) - b| < \varepsilon + \varepsilon = |a - b|,$$

a contradiction. Therefore $a = b$. $\square$

2.10. 01.10.2026.

2.10.1. *Further examples of function limits.*

Example 2.68 (Constant functions). If $f(x) = a$ for every $x \in \mathbb{R}$, then $\lim_{x \rightarrow x_0} f(x) = a$ for every $x_0 \in \mathbb{R}$. Indeed, given $\varepsilon > 0$, choose $\delta = 1$. Then $|f(x) - a| = 0 < \varepsilon$ whenever $0 < |x - x_0| < \delta$.

Example 2.69. For $f(x) = x^3$, we have $\lim_{x \rightarrow 2} f(x) = 8$.

Proof. If $|x - 2| < 1$, then $1 < x < 3$, so

$$|x^2 + 2x + 4| < 9 + 6 + 4 = 19.$$

Given $\varepsilon > 0$, choose $\delta = \min\{1, \varepsilon/19\}$. For $0 < |x - 2| < \delta$, factor the difference to obtain

$$|x^3 - 8| = |x - 2||x^2 + 2x + 4| < 19|x - 2| < 19\delta \leq \varepsilon.$$

$\square$

2.10.2. *Polynomial division.* Polynomial division lets us factor differences such as $P(x) - P(a)$, which is useful in direct proofs of limits.

Theorem 2.70 (Polynomial division). *Let P, D be polynomials with real coefficients, with $D \neq 0$. There are unique polynomials Q, R such that*

$$P = DQ + R,$$

where either $R = 0$ or $\deg R < \deg D$. We call Q the quotient and R the remainder.

Proof. Start with $Q = 0$ and $R = P$. If $R = 0$ or $\deg R < \deg D$, we are done. Otherwise, write the leading terms of R and D as ax^m and bx^d , respectively. Since $b \neq 0$ and $m \geq d$, replace

$$Q \text{ by } Q + \frac{a}{b}x^{m-d}, \quad R \text{ by } R - \frac{a}{b}x^{m-d}D.$$

The identity $P = DQ + R$ is preserved, and the leading term of R cancels. Thus the remainder becomes zero or its degree strictly decreases. After finitely many steps the required degree condition holds.

For uniqueness, suppose $DQ + R = D\tilde{Q} + \tilde{R}$, with both remainders satisfying the degree condition. Then

$$D(Q - \tilde{Q}) = \tilde{R} - R.$$

If $Q - \tilde{Q} \neq 0$, the left side has degree at least $\deg D$, while the right side is zero or has smaller degree. This is impossible. Hence $Q = \tilde{Q}$ and $R = \tilde{R}$. $\square$

Corollary 2.71 (Remainder and factor theorems). *The remainder when dividing $P(x)$ by $x - a$ is $P(a)$. In particular, $x - a$ is a factor of $P(x)$ if and only if $P(a) = 0$.*

Proof. Polynomial division gives $P(x) = (x - a)Q(x) + r$ with constant remainder r . Substituting $x = a$ gives $r = P(a)$. Thus the remainder vanishes exactly when $P(a) = 0$. $\square$

To divide by hand, arrange terms in descending powers, insert zero coefficients for missing powers, and repeatedly divide the leading terms, multiply the divisor, and subtract.

Example 2.72 (Cancelling a factor in a limit). Consider the limit

$$\lim_{x \rightarrow 2} \frac{x^3 + 5x^2 - 2x - 24}{x - 2}.$$

Both numerator and denominator tend to zero, so direct use of the quotient law is not possible. Divide $P(x) = x^3 + 5x^2 - 2x - 24$ by $x - 2$. The successive quotient terms are x^2 , $7x$, and 12 :

$$\begin{aligned} P(x) - x^2(x - 2) &= 7x^2 - 2x - 24, \\ (7x^2 - 2x - 24) - 7x(x - 2) &= 12x - 24, \\ (12x - 24) - 12(x - 2) &= 0. \end{aligned}$$

Therefore

$$x^3 + 5x^2 - 2x - 24 = (x - 2)(x^2 + 7x + 12).$$

For $x \neq 2$, we can cancel $x - 2$, giving

$$\frac{x^3 + 5x^2 - 2x - 24}{x - 2} = x^2 + 7x + 12.$$

Thus the desired limit is $4 + 14 + 12 = 30$. A direct verification uses

$$|x^2 + 7x + 12 - 30| = |x - 2||x + 9|.$$

If $|x - 2| < 1$, then $|x + 9| < 12$. Given $\varepsilon > 0$, take $\delta = \min\{1, \varepsilon/12\}$. For $0 < |x - 2| < \delta$, the difference from 30 is less than $12\delta \leq \varepsilon$. The quotient need not be defined at $x = 2$, because the limit uses only nearby points.

Example 2.73 (Using a remainder to simplify a limit). Consider

$$\lim_{x \rightarrow 1} \frac{x^3 - 1}{x^2 - 1}.$$

Again, both numerator and denominator tend to zero. Dividing $x^3 - 1$ by $x^2 - 1$ gives quotient x and remainder $x - 1$, since

$$(x^3 - 1) - x(x^2 - 1) = x - 1.$$

The remainder has degree $1 < 2$, so division is complete. For $x \neq \pm 1$, factor the denominator and cancel:

$$\frac{x^3 - 1}{x^2 - 1} = x + \frac{x - 1}{x^2 - 1} = x + \frac{x - 1}{(x - 1)(x + 1)} = x + \frac{1}{x + 1}.$$

This suggests the limit $1 + 1/2 = 3/2$. To verify it directly, suppose $0 < |x - 1| < 1/2$. Then $x + 1 > 3/2$, and

$$\begin{aligned} \left| \frac{x^3 - 1}{x^2 - 1} - \frac{3}{2} \right| &= \left| (x - 1) + \frac{1}{x + 1} - \frac{1}{2} \right| \\ &\leq |x - 1| + \frac{|x - 1|}{2|x + 1|} \\ &< 2|x - 1|. \end{aligned}$$

Given $\varepsilon > 0$, take $\delta = \min\{1/2, \varepsilon/2\}$. The displayed estimate is then less than $2\delta \leq \varepsilon$, proving the limit.

Remark 2.74. In both examples, division exposes a common factor that can be cancelled for $x \neq x_0$. Functions that agree throughout a punctured neighbourhood of x_0 have the same limit there: their defining ε - δ estimates are identical after shrinking δ to stay in that neighbourhood.

2.10.3. Limits using factorisation and local estimates.

Example 2.75. For $f(x) = x^3 + 5x^2 - 2x - 24$, we have $\lim_{x \rightarrow 2} f(x) = 0$.

Proof. By example 2.72, $f(x) = (x - 2)(x^2 + 7x + 12)$. If $|x - 2| < 1$, then $1 < x < 3$, and hence

$$|x^2 + 7x + 12| < 9 + 21 + 12 = 42.$$

Given $\varepsilon > 0$, choose $\delta = \min\{1, \varepsilon/42\}$. If $0 < |x - 2| < \delta$, then

$$|f(x) - 0| < 42|x - 2| < 42\delta \leq \varepsilon.$$

□

Example 2.76 (A piecewise function). Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} 2|x|, & x < 0, \\ 7, & x = 0, \\ x^2, & x > 0. \end{cases}$$

Then $\lim_{x \rightarrow 0} f(x) = 0$, even though $f(0) = 7$.

Proof. Given $\varepsilon > 0$, take $\delta = \min\{1, \varepsilon/2\}$. If $-\delta < x < 0$, then $|f(x)| = 2|x| < 2\delta \leq \varepsilon$. If $0 < x < \delta$, then $x < 1$, so $|f(x)| = x^2 < x < \delta \leq \varepsilon/2 < \varepsilon$. These cases cover $0 < |x| < \delta$. □

Example 2.77 (Keeping a denominator away from zero). Let $f : (0, 1) \rightarrow \mathbb{R}$, $f(x) = 7/x$. Then $\lim_{x \rightarrow 1/2} f(x) = 14$.

Proof. We have

$$\left| \frac{7}{x} - 14 \right| = \frac{14}{|x|} \left| x - \frac{1}{2} \right|.$$

If $|x - 1/2| < 1/4$, then $x > 1/4$, so $14/|x| < 56$. Given $\varepsilon > 0$, choose $\delta = \min\{1/4, \varepsilon/56\}$. Then

$$0 < \left| x - \frac{1}{2} \right| < \delta \implies |f(x) - 14| < 56 \left| x - \frac{1}{2} \right| < 56\delta \leq \varepsilon.$$

□

2.10.4. *A necessary Cauchy condition for function limits.* In the following results, domains contain a punctured neighbourhood of x_0 : for some $r > 0$, $(x_0 - r, x_0 + r) \setminus \{x_0\} \subseteq A$, as in the previous lecture's definition. All points at which a function is evaluated are understood to belong to its domain.

Theorem 2.78. If $\lim_{x \rightarrow x_0} f(x) = L \in \mathbb{R}$, then for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$|f(x) - f(y)| < \varepsilon$$

whenever $0 < |x - x_0| < \delta$ and $0 < |y - x_0| < \delta$.

Proof. Choose $\delta > 0$ so that $|f(t) - L| < \varepsilon/2$ whenever $0 < |t - x_0| < \delta$. For the stated x, y ,

$$|f(x) - f(y)| \leq |f(x) - L| + |f(y) - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

□

Example 2.79 (A jump prevents a limit). Let $f(x) = 1$ for $x \leq 0$, and $f(x) = 2$ for $x > 0$. The limit at 0 does not exist. Indeed, fix $\varepsilon = 1/2$. For every $\delta > 0$, the points $x = \delta/2$ and $y = -\delta/2$ satisfy $0 < |x|, |y| < \delta$, but

$$|f(x) - f(y)| = 1 > \varepsilon.$$

This contradicts the necessary condition in theorem 2.78.

2.10.5. The sequential criterion for function limits.

Theorem 2.80 (Sequential criterion). *Let $f : A \rightarrow \mathbb{R}$ and $m \in \mathbb{R}$. The following are equivalent:*

- (a) $\lim_{x \rightarrow x_0} f(x) = m$.
- (b) For every sequence (y_n) in $A \setminus \{x_0\}$ with $y_n \rightarrow x_0$, we have $f(y_n) \rightarrow m$.

Proof. Suppose (a) holds, and let $y_n \rightarrow x_0$ with $y_n \in A \setminus \{x_0\}$. Given $\varepsilon > 0$, choose $\delta > 0$ such that

$$0 < |x - x_0| < \delta \implies |f(x) - m| < \varepsilon.$$

There is $N \in \mathbb{N}$ such that $|y_n - x_0| < \delta$ for $n > N$. Since $y_n \neq x_0$, it follows that $|f(y_n) - m| < \varepsilon$ for $n > N$. Thus (b) holds.

Conversely, suppose (b) holds but (a) fails. The negation of (a) says that there exists $\varepsilon_0 > 0$ such that for every $\delta > 0$ we can find $y \in A$ satisfying

$$0 < |y - x_0| < \delta, \quad |f(y) - m| \geq \varepsilon_0.$$

For each $n \in \mathbb{N}$, choose such a point y_n with $\delta = \min\{r, 1/(n+1)\}$. Then $y_n \neq x_0$ and $|y_n - x_0| < 1/(n+1)$, so $y_n \rightarrow x_0$. But $|f(y_n) - m| \geq \varepsilon_0$ for every n , contradicting (b). Therefore (a) holds. $\square$

Remark 2.81. In the converse proof, we negate the statement that the limit equals the specified number m . This includes both a nonexistent limit and a limit different from m . The restriction $y_n \neq x_0$ is essential because $f(x_0)$ need not equal the limit.

2.10.6. Local boundedness and preservation of bounds.

Theorem 2.82 (Local boundedness). *If $\lim_{x \rightarrow x_0} f(x) = L \in \mathbb{R}$, there are $M, \delta > 0$ such that $|f(x)| < M$ whenever $0 < |x - x_0| < \delta$.*

Proof. Choose $\delta > 0$ using $\varepsilon = 1$ in the limit definition. Then

$$|f(x)| \leq |f(x) - L| + |L| < 1 + |L|$$

whenever $0 < |x - x_0| < \delta$. Take $M = 1 + |L| > 0$. $\square$

Theorem 2.83 (Preservation of bounds). *Suppose $\lim_{x \rightarrow x_0} f(x) = L \in \mathbb{R}$. If $f(x) \leq M$ throughout some punctured neighbourhood of x_0 , then $L \leq M$. If $f(x) \geq m$ throughout such a neighbourhood, then $L \geq m$. In particular, these conclusions hold if the bounds hold on all of A .*

Proof. For the upper bound, suppose $L > M$ and choose $\varepsilon = L - M > 0$. The limit definition gives $\delta > 0$ such that

$$f(x) > L - \varepsilon = M \quad (0 < |x - x_0| < \delta).$$

Shrink δ if necessary so that this neighbourhood lies in A and the assumed bound holds there. Taking $x = x_0 + \delta/2$ contradicts $f(x) \leq M$. Hence $L \leq M$.

For the lower bound, suppose $L < m$ and take $\varepsilon = m - L > 0$. For all sufficiently close $x \neq x_0$,

$$f(x) < L + \varepsilon = m.$$

Choosing such a point in the neighbourhood where $f(x) \geq m$ gives a contradiction. Thus $L \geq m$. $\square$

2.10.7. *Arithmetic laws for function limits.*

Theorem 2.84. *Let $f, g : A \rightarrow \mathbb{R}$, with $\lim_{x \rightarrow x_0} f(x) = a$ and $\lim_{x \rightarrow x_0} g(x) = b$. Then, for every $c \in \mathbb{R}$,*

$$\begin{aligned}\lim_{x \rightarrow x_0} (f(x) + g(x)) &= a + b, \\ \lim_{x \rightarrow x_0} cf(x) &= ca, \\ \lim_{x \rightarrow x_0} f(x)g(x) &= ab.\end{aligned}$$

If $b \neq 0$, then $g(x) \neq 0$ in some punctured neighbourhood of x_0 , and

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \frac{a}{b}.$$

The quotient is defined on $\{x \in A : g(x) \neq 0\}$.

Proof. Let $\varepsilon > 0$. All estimates below apply to points in a sufficiently small punctured neighbourhood of x_0 .

Sum. Choose $\delta_f, \delta_g > 0$ so that $|f(x) - a| < \varepsilon/2$ for $0 < |x - x_0| < \delta_f$, and $|g(x) - b| < \varepsilon/2$ for $0 < |x - x_0| < \delta_g$. With $\delta = \min\{\delta_f, \delta_g\}$,

$$|f(x) + g(x) - (a + b)| \leq |f(x) - a| + |g(x) - b| < \varepsilon.$$

Scalar multiple. If $c = 0$, the result follows from the constant-function example. If $c \neq 0$, choose $\delta > 0$ so that $|f(x) - a| < \varepsilon/|c|$. Then

$$|cf(x) - ca| = |c||f(x) - a| < \varepsilon.$$

Product. By theorem 2.82, there are $M > 0$ and $\delta_0 > 0$ such that $|f(x)| < M$ when $0 < |x - x_0| < \delta_0$. Choose $\delta_1, \delta_2 > 0$ to ensure respectively that

$$|g(x) - b| < \frac{\varepsilon}{2M}, \quad |f(x) - a| < \frac{\varepsilon}{2(|b| + 1)}.$$

For $0 < |x - x_0| < \min\{\delta_0, \delta_1, \delta_2\}$,

$$\begin{aligned}|f(x)g(x) - ab| &= |f(x)(g(x) - b) + b(f(x) - a)| \\ &\leq |f(x)||g(x) - b| + |b||f(x) - a| \\ &< \frac{\varepsilon}{2} + \frac{|b|\varepsilon}{2(|b| + 1)} < \varepsilon.\end{aligned}$$

Quotient. Assume $b \neq 0$. First choose $\delta_0 > 0$ such that $|g(x) - b| < |b|/2$. The reverse triangle inequality gives

$$|g(x)| \geq |b| - |g(x) - b| > \frac{|b|}{2} > 0.$$

Thus the quotient is defined near x_0 . To prove $1/g(x) \rightarrow 1/b$, choose $\delta_1 > 0$ so that $|g(x) - b| < \varepsilon|b|^2/2$. For $0 < |x - x_0| < \min\{\delta_0, \delta_1\}$,

$$\left| \frac{1}{g(x)} - \frac{1}{b} \right| = \frac{|g(x) - b|}{|g(x)||b|} \leq \frac{2|g(x) - b|}{|b|^2} < \varepsilon.$$

The product law applied to f and $1/g$ now yields $f(x)/g(x) \rightarrow a/b$. $\square$

Corollary 2.85 (Polynomial and rational limits). *For every real polynomial P and every $x_0 \in \mathbb{R}$, $\lim_{x \rightarrow x_0} P(x) = P(x_0)$. If Q is another polynomial with $Q(x_0) \neq 0$, then*

$$\lim_{x \rightarrow x_0} \frac{P(x)}{Q(x)} = \frac{P(x_0)}{Q(x_0)}.$$

Proof. The identity function has limit x_0 at x_0 : choose $\delta = \varepsilon$. Repeated use of the product law gives $x^k \rightarrow x_0^k$ for each positive integer k . Constant functions, scalar multiples, and finite sums then give the polynomial statement. The rational statement follows from the quotient law. $\square$

2.11. 06.10.2026.

2.11.1. *Infinite limits at a finite point.* As before, assume that the domain of $f : A \rightarrow \mathbb{R}$ contains a punctured neighbourhood of $x_0 \in \mathbb{R}$.

Definition 2.86. We write $\lim_{x \rightarrow x_0} f(x) = +\infty$ if for every $M \in \mathbb{R}$ there is $\delta > 0$ such that

$$0 < |x - x_0| < \delta \implies f(x) > M.$$

Similarly, $\lim_{x \rightarrow x_0} f(x) = -\infty$ if for every $M \in \mathbb{R}$ there is $\delta > 0$ such that

$$0 < |x - x_0| < \delta \implies f(x) < M.$$

Example 2.87. For $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$, $f(x) = 1/|x|$, we have $\lim_{x \rightarrow 0} f(x) = +\infty$. If $M \leq 0$, every $x \neq 0$ satisfies $f(x) > M$. If $M > 0$, choose $\delta = 1/M$. Then $0 < |x| < \delta$ implies $f(x) = 1/|x| > 1/\delta = M$.

2.11.2. *One-sided limits.* For the following definitions, the domain A need only contain the corresponding one-sided interval near x_0 .

Definition 2.88. The function $f : A \rightarrow \mathbb{R}$ has *left-hand limit* $L \in \mathbb{R}$ at x_0 , written $\lim_{x \rightarrow x_0^-} f(x) = L$, if for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$x_0 - \delta < x < x_0 \implies |f(x) - L| < \varepsilon.$$

The *right-hand limit* $\lim_{x \rightarrow x_0^+} f(x) = L$ is defined by requiring

$$x_0 < x < x_0 + \delta \implies |f(x) - L| < \varepsilon.$$

The left-hand limit is also denoted by $\lim_{x \rightarrow x_0-} f(x)$, and similarly on the right.

Example 2.89. Let

$$f(x) = \begin{cases} 1 - x, & x \leq 0, \\ 2x - 1, & x > 0. \end{cases}$$

For $x < 0$, $|f(x) - 1| = |x|$, so choosing $\delta = \varepsilon$ proves $\lim_{x \rightarrow 0^-} f(x) = 1$. For $x > 0$, $|f(x) - (-1)| = 2|x|$, so choosing $\delta = \varepsilon/2$ proves $\lim_{x \rightarrow 0^+} f(x) = -1$. The two-sided limit at 0 therefore does not exist.

Theorem 2.90. Suppose A contains a punctured neighbourhood of x_0 . Then $\lim_{x \rightarrow x_0} f(x) = L \in \mathbb{R}$ if and only if

$$\lim_{x \rightarrow x_0^-} f(x) = \lim_{x \rightarrow x_0^+} f(x) = L.$$

Proof. If the two-sided limit is L , its δ for a given $\varepsilon > 0$ works on each side of x_0 .

Conversely, suppose both one-sided limits equal L . Given $\varepsilon > 0$, choose δ_- and δ_+ from the respective definitions, and put $\delta = \min\{\delta_-, \delta_+\}$. If $0 < |x - x_0| < \delta$, then x is either in $(x_0 - \delta_-, x_0)$ or in $(x_0, x_0 + \delta_+)$. In either case $|f(x) - L| < \varepsilon$. $\square$

Example 2.91. For $f : [0, \infty) \rightarrow \mathbb{R}$, $f(x) = \sqrt{x}$, we have $\lim_{x \rightarrow 0^+} f(x) = 0$. Given $\varepsilon > 0$, take $\delta = \varepsilon^2$. Then $0 < x < \delta$ implies $|f(x) - 0| = \sqrt{x} < \sqrt{\delta} = \varepsilon$. Here only the right-hand limit is defined, since the domain has no points immediately to the left of 0.

2.11.3. *A necessary Cauchy condition for one-sided limits.*

Theorem 2.92. *Let $(x_0, x_0 + r) \subseteq A$ for some $r > 0$, and suppose $\lim_{x \rightarrow x_0^+} f(x) = L \in \mathbb{R}$. Then for every $\varepsilon > 0$ there is $\delta > 0$ such that*

$$|f(x) - f(y)| < \varepsilon \quad \text{whenever } x, y \in (x_0, x_0 + \delta).$$

The analogous statement holds for a left-hand limit.

Proof. Choose $0 < \delta \leq r$ so that $|f(t) - L| < \varepsilon/2$ for all $t \in (x_0, x_0 + \delta)$. Then, for any x, y in this interval,

$$|f(x) - f(y)| \leq |f(x) - L| + |f(y) - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

The left-hand case is identical. □